Rahul Natha

Mechanical & Computational Engineer – R&D
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Summary

Mechanical Engineer with 5 years of experience spanning electromechanical systems, engineering analysis, computational engineering, and design automation. Developed analytical methodologies, MATLAB-based engineering tools, parametric CAD frameworks, and experimentally validated mechanical systems. Focused on computational engineering, engineering optimisation, and simulation-driven product development.

Education

Bachelor of Technology in Mechanical Engineering — 08/2016 – 09/2020

Jawaharlal Nehru Technological University Hyderabad — Hyderabad, India

- CMR College of Engineering & Technology
  Grade: 8.91/10.00 CGPA; Awarded the Gold Medal

  Project: Applied Taguchi methods, ANOVA, regression analysis, and empirical modelling to optimise Wire EDM machining performance of Maraging Steel 250.

Upper Secondary Education — 06/2004 – 03/2016

Telangana State Board — Hyderabad, India

- Sri Sanjeevni Junior College
  Grade: 71.5%; Higher Secondary School up to Class 12; Stream: Mathematics, Physics, & Chemistry
- St. Augustine High School
  Grade: 8.8/10.0 CGPA; Secondary School up to Class 10

Experience

Senior, Junior Research Fellow — 03/2022 – 03/2026

Defence Research & Development Organisation — Hyderabad, India

Designed and developed mechanical architectures for closed-loop electromechanical actuation systems at Research Centre Imarat, a DRDO laboratory, emphasising high-precision motion control, structural integrity, and experimental-to-simulation data correlation for aerodynamic control surface applications.

- Designed and released complete manufacturing documentation for a closed-loop electromechanical actuator comprising gear trains, torque shafts, bearings, encoder and motor interfaces, PCB integration features, and housing assemblies, supporting prototype fabrication and system integration.
- Led structural verification of a precision electromechanical actuator assembly using ANSYS Mechanical, defining load cases, contact interactions, and acceptance criteria that validated all critical components prior to prototype manufacture, ensuring a factor of safety ≥ 1.5 across all operating envelopes.
- Designed high-precision gear train assemblies achieving micron-level alignment, maintaining critical centre-to-centre bearing distances to a ±10 micrometre tolerance in compound gear arrangements to support low-backlash motion control.
- Developed MATLAB-based design automation tools for non-standard rectangular cross-section helical torsion springs, transforming a previously manual engineering workflow into a repeatable computational design process.
- Developed MATLAB-driven parametric spring design and CAD automation templates that reduced engineering drawing generation time by over 90%, enabling same-day design iterations and accelerating prototype release schedules.
- Correlated analytical, FEA, and experimental results for custom torsion spring designs, achieving prediction accuracy within 5% of measured performance and validating the developed design methodology.

Mechanical Engineer
Cyient

05/2021 – 03/2022
Hyderabad, India

- Developed Excel-driven XML generation tools for Arbortext technical publications, reducing manual effort by over 90% in recurring schematic legend authoring workflows for a Fortune 100 agricultural equipment manufacturer.
- Developed technical illustrations and engineering documentation for multinational aerospace, rail transportation, and industrial equipment manufacturers, applying industry-standard drawing practices, callout conventions, and publication requirements.
- Converted legacy 2D engineering documentation into parametric 3D CAD models, improving geometric traceability and supporting downstream documentation workflows.

Industrial Design Intern
MechRise Solutions

05/2019 – 07/2019
Hyderabad, India

- Gained hands-on mechanical design experience with solid modelling tools and foundational fabrication methodologies.
- Studied procurement and production fundamentals through structured training, focusing on cost and industrial design.

Skills

- **Computational Engineering**: MATLAB, Python, Engineering Computational Scripts, Algorithmic Design Automation, Numerical Modelling of Mechanical Components, Parametric CAD Automation, Advanced Excel Array Frameworks.
- **Simulation & Analysis**: Finite Element Analysis (ANSYS Mechanical: Static Structural), Mechanics of Solids, Precision Gear Mechanics (KISSsoft), Bearing Selection & Analysis, Actuator Kinematics, Finite Element Model Correlation (FEA-to-Test).
- **CAD & Manufacturing**: Siemens NX, SOLIDWORKS Premium, CATIA, Creo Parametric, Autodesk Fusion 360 (Certified User), Parametric Template Development, Top-Down Assembly Modelling, GD&T (ASME Y14.5), Tolerance Stack-Up Analysis, DFM/DFA.
- **Optimisation & Testing**: Design of Experiments (DoE), ANOVA, Minitab, Empirical Data Correlation, Environmental Qualification Testing, Fault Isolation Protocol Development.
- **Engineering Methods**: DFMEA, VAVE, System Cost Optimisation.
- **Languages**: English [C1, IELTS Academic 7.0 – Valid until 24 May 2027], Deutsch [A1], Telugu, Hindi.

Certifications & Activities

- Linear and Non-Linear Finite Element Analysis with Programming; Organised by BITS Pilani, Hyderabad Campus (featuring Prof. J. N. Reddy, Texas A&M University). 2023
- ANSYS Mechanical Training; Issued by 3D Engineering Services. 2023
- Siemens NX Mastery: Advanced Design & Applications; Issued by Siemens. 2026 (ongoing)
- SOLIDWORKS 3D CAD for Education; Issued by Dassault Systèmes. 2025
- Fusion (Fusion 360) Certified User; Issued by Autodesk. 2019
- Robotics & Industrial Automation, with hands-on training on KUKA KR-16 Path Programming and Calibration; Issued by CIT KUKA Training Centre; @ Chennai, India. 2018
- CNC Milling; Issued by National Skill Training Institute; @ Hyderabad, India. 2018
- McKinsey.org Forward Program; Issued by McKinsey.org. 2026
- Delegate at the 13th Asia Youth International Model United Nations (AYIMUN); Organised by International Global Network; @ Kuala Lumpur, Malaysia. 2024
- German Language Training; Organised by NSDC International; @ Skill India International Centre, Hyderabad, India. 2026 (ongoing)